The Echo Chamber Effect Resounds on Financial Markets: A Social Media Alert System for Meme Stocks

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Abstract

The short squeeze of Gamestop (GME) has revealed to the world how retail investors, pooling through social media, can severely impact financial markets. In this paper, we devise an early warning signal to detect suspicious users' social network activity, which might affect the financial market stability. We apply our approach to Reddit data, selecting both meme and non-meme stocks as case studies. The alert system is structured in two stages; the first is based on extraordinary activity on the social network, while the second aims at identifying whether the movement seeks to coordinate the users to a bulk action. We run an event study analysis to see the reaction of the financial markets when the alert system catches social network turmoil.

Keywords: Market manipulation; Social network analysis; Alert system; Event study; Reddit.

JEL codes: G14; G18; G41; 039.

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1 Introduction

Retail investors have usually been considered noise traders in the financial and market microstructure literature: their choices are not driven by the knowledge of the fundamentals of a stock or any sophisticated analysis of the market, but they are guided primarily by their emotional and irrational beliefs (see Black (1986)). Noise traders' market impact has always been considered negligible by the "neoclassical finance" compared to the influence of institutional investors, such as investment banks and hedge funds.[1]. However, this picture of nonthreatening amateur investors looks outdated: the progressive diffusion of social media combined with low-cost trading platforms are making investment strategies more and more widespread. The impact they have on the financial market is anything but negligible, as highlighted by some recent events. The most striking episode is the short-squeeze that some retail investors triggered on GameStop stock (ticker symbol on NYSE: GME) by coordinating themselves on Reddit, a micro-blogging social network (Anand and Pathak, 2021).

Reddit is a website[2] composed of user-generated content and related discussions. The site's content is divided into forums and communities known as "subreddits", which deal with specific topics of interest. As a network of communities, Reddit's core content consists of posts (submissions) from users. Users can comment on others' posts to continue the conversation, and they can collect positive or negative votes (score). The number of upvotes or downvotes determines the posts' visibility on the site: the more popular the content is, the higher the number of people it is displayed.

One of the most popular and active subreddit is *r/WallStreetBets*, a community focused on financial markets and stock trading. In this community, users boast very aggressive trading strategies and what they did at the end of January 2021 with GameStop is no exception. GameStop (GME) is the world's largest American retailer of video games and accessories. The market for physical video games started its decay in 2017 due to the downloadable version of many games and services offered by the main consoles. GameStop has experienced a sharp decline in sales which has led to a share price drop in the financial market. The COVID-19 pandemic was a severe beating for the company, and its fundamentals faced another shock. The stock price decline led many institutional

[1]For a detailed review on noise traders, see among others Ramiah et al. (2015)

[2]In the top-10 of the most visited websites in the US according to `https://www.alexa.com/topsites/countries/US` and `https://www.semrush.com/blog/most-visited-websites/`

investors to sell the stock short. Conversely, some retail investors, considering the stock undervalued, went against the trend. In January 2021, a coordinated effort orchestrated by the community of the subreddit

r/WallStreetBets surged the price of GME (US Securities and Exchange Commission, 2021).

Apart from what happened in the financial market, the squeeze of the price and the consequent losses faced by the short-sellers, the high volatility of traded volumes and the liquidity issues, the most intriguing part of this episode is figuring out how an apparently harmless group of noise traders were able to provoke such a substantial effect on a market usually dominated by the big players.

Stylizing the phenomenon can be helpful to detect potential anomalies based on indicators of agents' coordination. This is relevant from the perspective of policy-making to prevent this kind of coordinated action from happening or at least to tackle the harmful effects.

Besides the specific case of GameStop, which is dramatic in terms of magnitude and subsequent effects, the interest in how social media networks impact the financial price formation of meme stocks is gaining growing attention (Pedersen, 2022a; Costola et al., 2021). Furthermore, an ever-increasing decentralization of the financial system and the ease to access it via user-friendly online trading platforms are potential destabilizers for the financial ecosystem[3]. The fintech (r)evolution and the capillary diffusion of the online social network, which allow us to receive up-to-date information as they happen, lay the foundations for a reinterpretation of market manipulation steering through social networks. Hence, there is an urgent need to develop an alert system to pinpoint episodes of potential misconduct in online social networks that can result in possible market manipulation. With misconduct behavior, in this paper, we refer to an inappropriate activity at the social network level that might potentially translate into market manipulation, a deliberate attempt to interfere with the free and fair operation of the market.

Although there is no regulation legislating on the relationship between social media coordination and the financial market, the U.S. securities and exchange commission (SEC) defines it as market manipulation[4] *"all the actions where someone artificially affects the*

[3]See for instance https://www.risk.net/investing/7810026/aqr-quant-on-the-network-effects-behind-gamestop-frenzy

[4]https://www.investor.gov/introduction-investing/investing-basics/glossary/market-manipulation

supply or demand for a security".

Social network variables on mass coordination are valuable tools for building nowcasting systems and scheduling real-time interventions to ensure stability. Our paper contributes to the extant literature by designing an alert system to detect potential misconducting behavior or suspicious activity on the social network to eventually prevent harmful coordination from creating instability in financial markets. The methodology relies on social listening and social network analysis to identify the red light activities to monitor. The rest of the paper proceeds as follows: Section 2 reviews the literature on the topic, Section 3 presents the data and their features, Section 4 describes the alert system we develop, and Section 5 introduces the empirical results. Finally, Section 6 concludes.

2 Literature Review

Digital and online social network revolutions are deeply affecting the functioning of financial markets. One manifestation of this revolution is the rising importance of retail traders. In classical market microstructure models (Glosten and Milgrom, 1985; Kyle, 1985), noise traders are considered as a residual category because of their randomness in the trades. Noise traders are usually ignored in the price formation process because their irrational impact on the market (which makes the price diverge from the fundamental value temporarily) is predominated and counterbalanced by rational agents on the market.

One of the first models to recognize the relevance of noise traders in the financial ecosystem is the one described by De Long et al. (1990) where the authors model them as irrational traders with erroneous stochastic beliefs whose *"unpredictability creates a risk in the price of an asset. As a result, prices can diverge significantly from fundamental values even without fundamental risk"*. This paper was highly enlightened and forward-looking as it perceived how the agents, so far targeted as irrelevant, are effectively impacting the asset price formation in particular circumstances.

Thanks to user-friendly, low-cost online trading platforms, the widespread diffusion of social media and the easiness of accessing financial markets have significantly transformed the market dynamics. As pointed out by Zheludev et al. (2014), the proliferation of the internet has improved our ability to access information in real-time, and in particular, the diffusion of social media allows us to get in contact with the moods, thoughts, and opinions of a large part of the world's traders in an aggregated and real-time manner. Many investigations aim to analyze the impact of social media on the financial markets both from a perspective of the volume of social activity, the search engine traffic, and the prevailing sentiment of the agents (Mao et al., 2012; Ruiz et al., 2012; Bordino et al., 2012; Preis et al., 2013). A branch of the literature focuses, using various techniques, on the impact of social media activity on the financial markets (Mao et al., 2012; Bordino et al., 2012; Ruiz et al., 2012; Preis et al., 2013) . Most of the available results in this direction are based on empirical extrapolation.

In Renault (2018), the author focuses on a specific type of market manipulation: the pump-and-dump scheme. He finds that abnormal activity on social media about a stock is associated with a sharp increase in volume and price on the day of the event, while the price presents a reversal in the following trading week. Pedersen (2022a) proposes a new model revolutionizing the vision of the so-called noise traders. He describes how investment strategies propagate in a social network affecting the market. Four typologies of investors are considered: besides the classic prototypes of rational short- (who tries to predict the sentiment changes among naive investors based on social network dynamics) and long-term investor (focused on the fundamental value of assets), it portrays a new type of agents: the 'fanatics' (investors with a stubborn view that can influence many people thanks to their popularity on social media) and naive investors (agents learning and relying on investment strategies proposed on social networks). The model explains the belief formation process on the social network and how it affects the fluctuations of prices and trading volumes on the financial market. Modern social media, such as Reddit, allows envisioning (and downloading) the data-generating process that leads to the coordination of the agents on the network and analyzes the underlying forces behind the event. To the extent of our knowledge, the literature on (social) network evolution and consequent impact on financial markets is still in its infancy, and there are very few works accounting for this issue (see Lucchini et al. 2022). Suppose social network activity does generate a force that drives the financial market dynamics. In such a case, the GameStop saga can be an exciting case study to understand the network indicators to monitor to prevent extreme phenomena like the one that happened at the end of January 2021. Hence, as recommended by Pedersen (2022b) in the conclusion of his paper, the availability of data from social media might open *"new research possibilities to test model's prediction using data on networks and market behavior"*.

Dim (2021) analyzes the implications on the financial markets of *"non-professional social media investment analysts"* that publish investment strategies shaping the views and actions of many retail investors. This study highlights how the interplay of social media and retail trading poses new challenges for financial markets and regulators, which requires a deeper understanding of belief formation on social media.

Our paper works towards defining some indicators and parameters of the social network to be monitored to detect extreme situations that might affect the financial market stability. Inspired by the setting proposed in Costola et al. (2021), our alert system has two consecutive red flags: if the first one, based on extraordinary activity on the social network, activates, we start monitoring the structure of the user social network.

In network analysis, distinguishing the various roles the users can play is crucial. Many works are devoted to this categorization (Ríos et al., 2019; Choi et al., 2015; Thukral et al., 2018). A special role is played by the influencers, aka the leaders, or to use a neologism proposed Pedersen (2022a), the "fanatics". We track the users' activity within the network and catch the leading players denoted by their ability to be central and vocal in the network proposing relevant contributions.[5]

Before delving into the contribution of our paper, we cross-reference all the works dealing with the GameStop case, the triggering factor of this piece of literature devoted to creating an alert system to prevent the dysfunctional social network activity from destabilizing the financial market. Investment recommendation (Bradley et al., 2021), social network activity volume and tone (obtained with sentiment analysis) (Long et al., 2021; Umar et al., 2021) influence the GME returns and traded volume on the financial market. Also, the Google trend researches with keywords related to the event are positively correlated with the financial GameStop performance (Klein, 2021; Vasileiou et al., 2021). Hasso et al. (2021) profile the agents participating in the frenzy and describe their average performance; they have proven to be relatively inexperienced and unsophisticated (Eaton et al., 2021). Eaton et al. (2021) also argue that a large portion of agents acting on

[5]A very recent contribution that uses applied network analysis to the GameStop short squeeze is developed by Mancini et al. (2022). The authors introduce a model that evaluates how the emergence of consensus lead to the GME short-squeeze in January 2021.

the subreddit *r/WallStreetBets* uses Robinhood as a trading app. Robinhood is a zero commission, no account minimum, and an easy-to-use interface trading app widely used among young investors. During the most turbulent period of GME frenzy, the trading app went down or malfunctioned several times, avoiding investors from acting on the financial market and losing the best moments to trade. Many were complaints about this malfunction reported on a website *DownDetector.com*[6], an online platform that provides users with information about the status of various websites and services based on user outage reports.

The disruptive effect would have been milder if only the financial market presented more substantial barriers to entry. Finally, Fusari et al. (2021) report as a case study the extreme event of GME, demonstrating that they can predict the bubble using a model based on options.

Against this background, we provide an event analysis based on the detection of potential users that might trigger a collective action on the financial markets. The methodology is based on a two-step analysis of a specific subreddit in the community. The first step is based on the identification of days featuring high volumes of activity on social media. The second step aims to detect potential influential users leading the community toward a particular (coordinated) action.

3 Data

Our analysis is run on selected NYSE-listed stocks in the period October 2020 - April 2022 for the following stocks (NYSE ticker is reported in parenthesis): GameStop (GME), American Multi-Cinema Entertainment (AMC), Tesla, Inc. (TSLA), Nokia (NOK), Apple Inc. (AAPL) and Microsoft Corporation (MSFT). For each stock, we download the financial daily data on price and trading volume and all the posts (submissions) and comments on the Wallstreetbets subreddit containing the ticker[7]. GME, AMC, TSLA and NOK are examples of meme stocks[8], meaning stock that gains popularity among retail investors through social media. As reported in US Securities and Exchange Commission

[6]https://downdetector.com/

[7]For data collected after June 2021 we download data from other subreddit, such as *r/WallStreetbetsnew, r/GME, r/AMC*. The reason behind the choice is that participation between (real) users in the Wallstreetbets community significantly dropped after a certain date.

[8]At this webpage you can find a constantly updated list of meme stocks, defined as tickers with most mentions in the Wallstreetbets subreddit.

(2021), a meme stock is characterized by a confluence of all these factors: large price moves, large volume changes, large short interest, frequent mentions on social media and significant coverage in the mainstream media. Conversely, AAPL and MSFT are two non-meme stocks to be used as controls.

In section 3.2 we describe how we downloaded and used the data from Reddit. At this webpage you can find our dataset with all the Reddit submissions containing the stock's tickers.

3.1 Market data

We download the time series with a daily resolution of price and traded number of shares (volume) from October 1^{st} 2020 to April 30^{th} 2022. For each stock we compute the daily returns:

$$R_{t,s} = \frac{P_{t,s} - P_{t-1,s}}{P_{t-1,s}} \tag{3.1}$$

where $P_{t,s}$ is the closing price of stock s on day t. The daily trading volume is denoted as $Vol_{t,s}$.

3.2 Reddit data

Rooting our analysis in the theoretical framework proposed by Pedersen (2022a), we design a model to study the interaction of agents on the social platform, evaluate their coordination effort, and quantify the impact of their common action on the financial market. Modern social media contain an outstanding informative potential related to the users' sentiment evolution and opinion formation. If properly squeezed, the information between users can be a forerunner for upcoming events.

Relying on the informative power of the social network and with the help of PRAW (an acronym for 'Python Reddit API Wrapper', a Python package that allows accessing Reddit's API), we downloaded all the posts containing as a keyword the ticker of the stock to investigate. We limit the data download to the subreddit *r/WallStreetBets* in the period from October 2020 to June 2021. After June 2021 we realized participation of real users in the community significantly decreased, therefore we extended the download adding other three subreddits: *r/WallStreetbetsnew, r/GME, r/AMC* (the last two stock-specific). In the end, we collect data for the sample October 2020-April 2022.

For each post, we obtained the related comment forests and attributes. We downloaded the data and run the analysis for the following tickers: GME, AMC, TSLA, NOK, AAPL, and MSFT. Precisely, we extracted every post (in Reddit jargon 'Submission') containing the ticker and its related comment tree.[9]

We are interested in the emerging collective phenomena when retail investors cooperate to determine a significant effect on the financial market. Therefore, we excluded from our analysis the messages generated automatically from the bots (that in our dataset are denoted with the tag 'moderator' in the variable `distinguished`). A conversation thread can be modeled as a directed tree $T^k_{t,s} = (M^k_{t,s}, C^k_{t,s})$. $M^k_{t,s}$ the set of nodes represented by the messages in the tree k (where the initial submission represents the root of the tree and the comments are the following-up branching) and $C^k_{t,s}$ is the set of edges, each of them connecting two messages linked by commenting activity. The direction of the edge points to the parent node to which the comment is addressed.

Figures 1, 2 report the number of daily posts (i.e. the number of trees) containing as a keyword the ticker in the title of each subgraph and related comments (i.e. the number of nodes excluding the initial submission). Each subgraph has a double scale y-axis; on the left (in red) is the scale measuring the absolute value of daily submissions, on the right (in blue) is the corresponding measure for the number of related comments. We separate two periods: 2020/10-2021/07 (Figure1 1) and 2021/07-2022/04 (Figures 2).[10] The first period is characterized by high activity for all the stocks. During the second period, the data volume is significantly lower. The plots in Figures 1 and 2 clearly show a sudden change in the behavior of Reddit users. Indeed, the GameStop short squeeze focused media attention on the meme-stocks phenomenon, with the financial authorities opening legal cases against some users. The Wallstreetbets community was under worldwide attention, discouraging the users in their activity on the public forum and potentially favoring new forms of coordination in private forums or anonymous messaging apps. Therefore the reduction in the movement affected the entire community. These changing behaviors pose many challenges in analyzing Reddit users and their impact on the financial markets.

[9] See the Appendix A for more details of downloaded data.

[10] In a preliminary version of the paper, we only focused on the first period of the analysis. We included the following period subsequently.

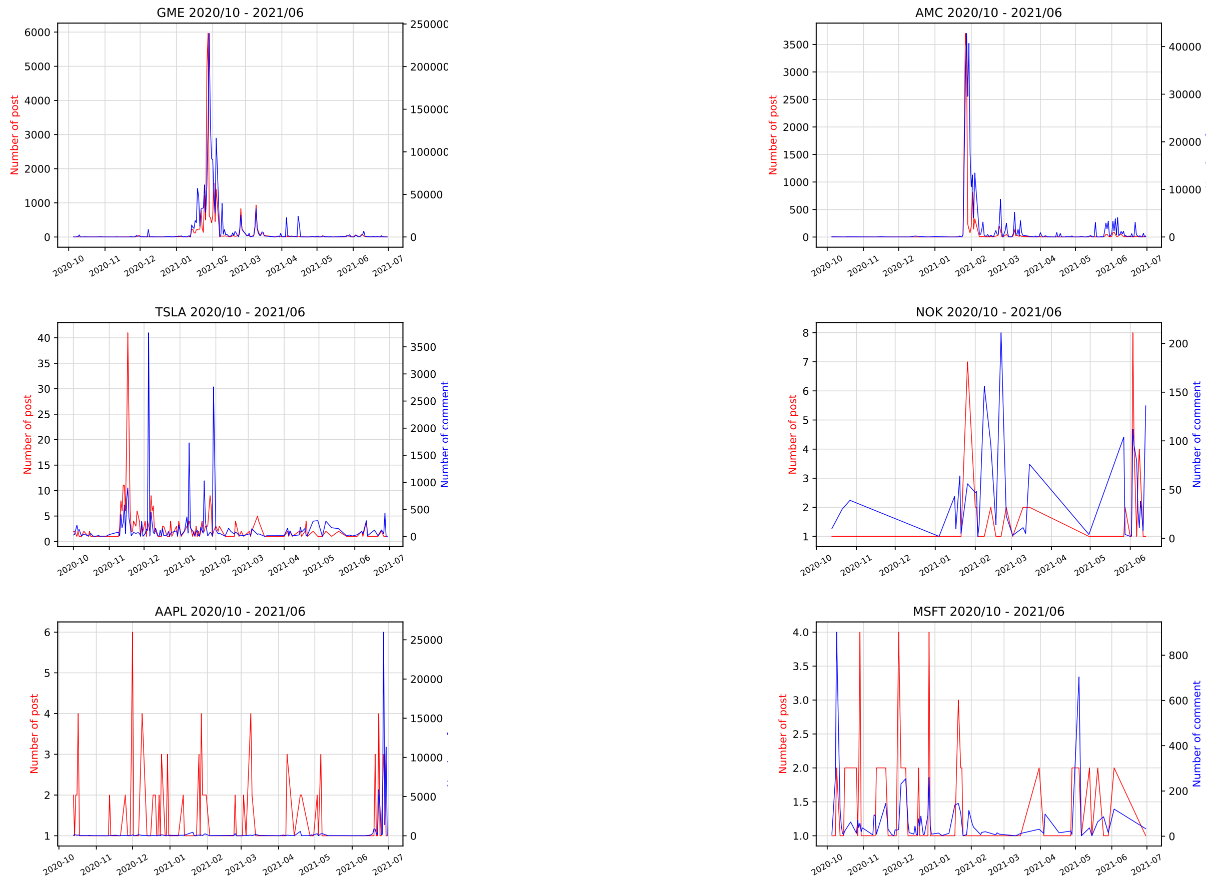

Figure 1: Time series of the daily number of posts (submissions) containing the word reported in the title of each subgraph (GME, AMC, TSLA, NOK, AAPL, MSFT) and related comments on the subreddit *r/Wallstreetbets* in the period October 2020 - June 2021. The post y-axis scale measure is on the left in red, while the corresponding scale for the comments is on the right of each graph and in blue.

3.2.1 Social network analysis

We extrapolate the daily agent interplay from the tree-level raw data by reconstructing their discussion interchanges. We define a network where the nodes represent the active users on the day t, and the edges are users' interactions through the comments: the network structure is outlined by $G_{t,s} = (N_{t,s}, E_{t,s})$. The set of active users on the day t for ticker s, $N_{t,s}$, represents the graph's nodes. The commenting activity establishes the links among them: $E_{t,s}$ is the set of directed edges from the user writing the comment to the author of the initial submission.

Following Ríos et al. (2019), we adopt a simplification in the user-network reconstruction (see Figure 3): the links always point to the author of the initial submission even if the comment is a reply to a second or higher-order comment. The ratio underlying this network reduction is the need to identify the users acting like hubs, gaining popu-

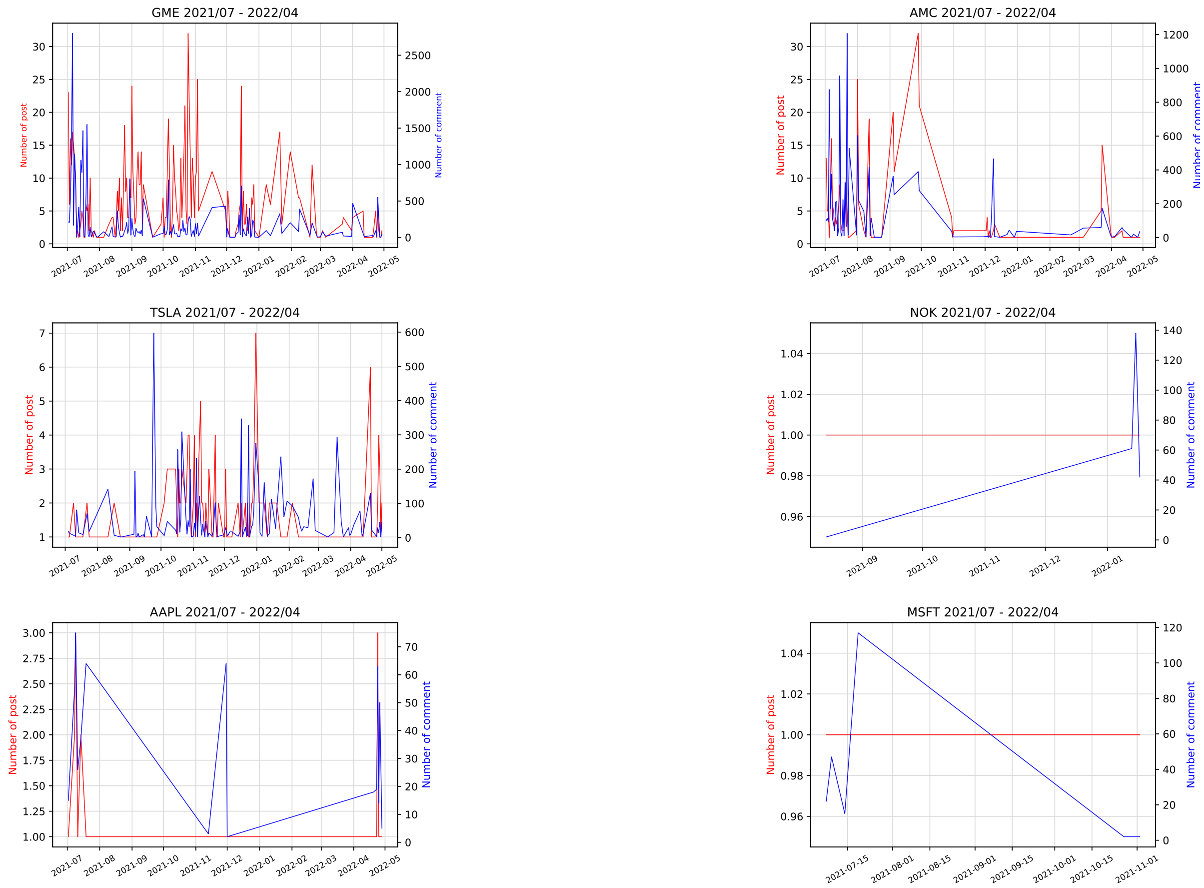

Figure 2: Time series of the daily number of posts (submissions) containing the word reported in the title of each subgraph (GME, AMC, TSLA, NOK, AAPL, MSFT) and related comments on the subreddit *r/Wallstreetbets* in the period July 2021 - April 2022. The post y-axis scale measure is on the left in red, while the corresponding scale for the comments is on the right of each graph and in blue.

larity, consensus, and driving a potentially impactful coordinated action. The simplified network representation we adopt is still a realistic description of the social dynamics: when scrolling the blog, the user first reads the main submission; then, if attracted by the topic, she will open the cascade of comments. Our reduction considers all the comments as first-level comments. This structure emphasizes the submitter that becomes the central actor of a thread and, eventually, the driver of a particular discussion. Figure 4 depicts an example of interaction among users. The network represents the interaction on Reddit on January 14st, 2021, based on submissions containing the ticker GME. There are 6 465 users (represented by the nodes) interacting on the platform throughout commenting activity (the 8 741 edges connecting them). The social network has a hub-and-spoke structure, with the colored users representing the hubs (i.e., central nodes in the network). In Appendix B we report similar networks for AMC, AAPL and MSFT.

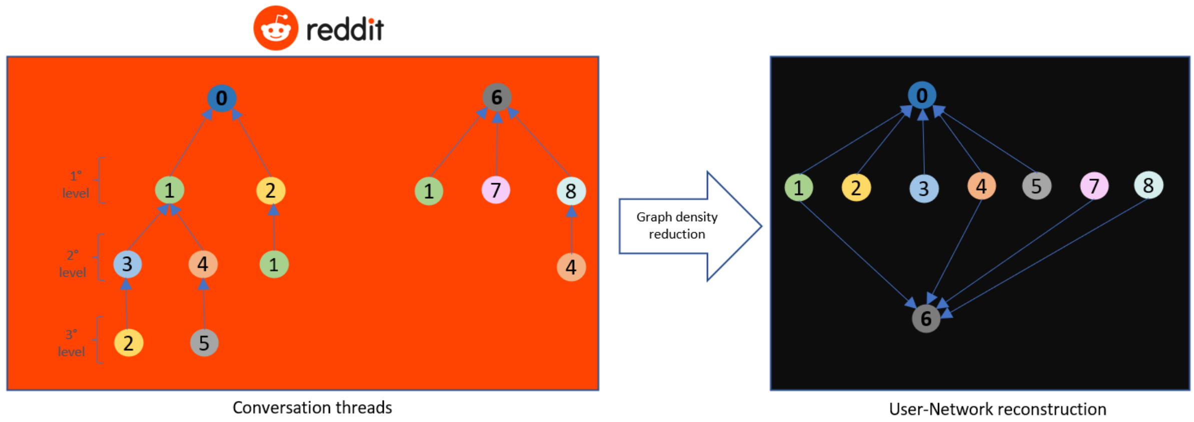

Figure 3: Example of user network reconstruction from two conversation threads. Starting from two conversation threads $T^k_{t,s} = (M^k_{t,s}, C^k_{t,s}), k = 1, 2$ stylized as comment trees (left-hand panel), we reconstruct the interaction among users $G_{t,s} = (N_{t,s}, E_{t,s})$ by reducing the density and the level of detail. We consider all the comments to be addressed to the submission creators (in the example the users labeled with 0 and 6), as all the comments were at the first level of the tree. We do not consider how many times a user interacts with another one: we consider whether a user comments to the submitter to streamline the social network.

4 Methodology

This section presents the backbone of our analysis. We describe the design and the functioning of the alert system to detect potential misconducting behavior at the social network level. Subsequently, we illustrate how we structure an event study analysis to check whether the alert system is capable of anticipating potential attempts of market manipulation.

4.1 Alert system

We devise an alert system based on social network-retrieved information. Cooperation among users can translate into a dangerous impact on financial market stability. Detecting potential misconduct behaviors and anticipating a coordinated action conducted on social media might be beneficial for the financial market well-functioning. The alert system is query-dependent, meaning that we have to instruct the system on which keyword we want to monitor. For the sake of practical usability, two consecutive stages compose the structure.

4.1.1 First stage

The first stage of the alert system consists of a screening of the days where the ticker-related activity on the social network is extensive compared to the previous days. We use the volume-related metrics to implement the first stage: we skim the days, identifying when ticker-related activity is extraordinary. Technically speaking, for every day and each ticker, we determine:

- The number of submissions citing the ticker in their content;
- The number of active users discussing that ticker;
- The overall[11] activity on the subreddit, both in terms of submissions and users.

We construct the following variables:

(1) Relative number of daily submissions: the ratio between the submissions citing ticker and the total submissions on the subreddit, to identify the portion of the activity on that topic during the day;

(2) Absolute number of daily submissions: the total number of daily submissions about that ticker, to identify the magnitude of the movement in absolute terms;

(3) Relative percentage change in the number of daily submissions: the percentage variation of the number of ticker-related submissions with respect to the previous day;

(4) Relative number of daily users: the ratio between the number of users citing ticker and total users on the subreddit, to identify the portion of the community discussing that topic during the day;

(5) Absolute number of daily users: computed as the total number of daily users citing that ticker, to identify the magnitude of the user-trend in absolute terms;

(6) Relative percentage change in the number of daily users: the percentage variation of the number of ticker-related users with respect to the previous day;

[11]Meaning the whole activity during the day on the subreddit *r/WallStreetBets* (or *r/WallStreetBetsnew, r/GME, r/AMC* for data collected after June 2021), not specifically related to a ticker.

The first stage of the alert system turns on when at least four of the above-listed variables exceed their threshold[12]. For the relative and the absolute number of daily submissions (users), variables (1), (2), (4), (5), the threshold for the day t is the mean value of the variable over the previous ten days plus one mean absolute deviation computed over the same period. We define the mean value of the variable as

$$\bar{V}_{t,s}^{(v)} = \frac{1}{I} \sum_{i=1}^{I} V_{t-i,s}^{(v)} \tag{4.1}$$

The mean absolute deviation as

$$MAD(V_{t,s}^{(v)}) = \frac{1}{I} \sum_{i=1}^{I} \left(V_{t-i,s}^{(v)} - \bar{V}_{t,s}^{(v)} \right) \tag{4.2}$$

Hence, the threshold is defined as

$$T_{t,s}^{(v)} = \bar{V}_{t,s}^{(v)} + MAD(V_{t,s}^{(v)}) \tag{4.3}$$

where v refers to the variables $j = 1, 2, 4, 5$, t indicates the day, s the ticker $s = GME, AMC, AAPL, MSFT$ and I is the length of the window over which we compute the threshold, in our case $I = 10$. For the relative percentage change in the number of daily submissions (users), variables (3) and (6), the threshold to overcome is 100%, hence when on the day t they double the value with respect to the previous day $t - 1$.

When the social network conditions determine the activation of the alert, we approach the situation in a prudential and conservative way: we keep monitoring the stock until the average number of active indicators over the previous three days is below three. In this way, we keep controlling the situation even if it is not exceptional compared to the earlier days, but it is still turbulent.

A single indicator (or variable) switches on when it overcomes its critical threshold defined in (4.3):

$$I_{V_{t,s}^{(v)}} = \begin{cases} 1, & \text{if } V_{t,s}^{(v)} > T_{t,s}^{(v)} \\ 0, & \text{otherwise} \end{cases} \tag{4.4}$$

[12]Notice that we set the number four as an apriori threshold to give a certain confidence to the alert.

The minimum alert-activation condition is:

$$\sum_{v=1}^{6} I_{V_{t,s}^{(v)}} \geq 4 \tag{4.5}$$

The alert remains on until the following condition verifies:

$$\frac{1}{3}\sum_{i=1}^{3}\sum_{v=1}^{6} I_{V_{t-i,s}^{(v)}} \leq 3 \tag{4.6}$$

Step one of the alert system detects the days when the activity is exceptional, calling for further controls on the network structure.

4.1.2 Second stage

The second stage of the alert system activates only for those days recognized as potentially critical in the first stage.

For each day selected by the first step, we reconstruct the network structure to model the interaction among the agents, $G_{t,s} = (N_{t,s}, E_{t,s})$ as described in the previous paragraph: the links always point to the author of the initial submission even if the comment is a reply to a second or higher-order comment.

We also implement other filters in the network modeling: creating a nowcasting alert system requires the tool to be quick and smartly devised, beyond the fact that it has to detect the bigwigs. Hence, the network reconstruction is made of only the users whose submission obtained a score above the median and with a cascade of at least ten comments.

We now move to the detection of the users acting like leaders, able to gain trustworthiness, popularity, prestige, and authority, essential features for a virtual user to lead a movement that can translate its effects on the real economy.

We rank the users according to their in-degree centrality (fraction of nodes its incoming edges are connected to) for each day in the subset of first-stage-detected days:

$$C_D(n_{t,s}^{(i)}) = \frac{\sum_j a_{t,s}^{ij}}{N_{t,s}} \tag{4.7}$$

The indicator identifies the ten authors with the highest relative incoming links: users able to attract a vast portion of the community. To test whether the detected authors are trending and critically acclaimed, for each day t, we define a set of the most critical users according to the algorithm of PageRank(Page et al., 1999). We consider the network in the window $[t-60, t-1]$ and identify the first twenty users who were standing out for the published content, constituting the set of influencers at time t. We finally check whether some of the top in-degree centrality authors on the day t belong to the influencers set over the past sixty days[13]. If the intersection between the two sets is not empty, the second stage of the alert system turns on.

The methodology narrows down the set of agents we monitor to avoid misconducting behaviors on the online social network and prevent repercussions on financial stability. The method aims to pin down the users who might manage suspicious movements by promoting extreme investment strategies masked by financial advice. From the perspective of macroprudential stability, a regulatory authority using our tool can quickly pinpoint the users to check to analyze their contents through textual analysis tools and eventually ban the profiles.

To more weight to days when few but catching influencers are active, we propose a 3-level warning alert system: the severity of the signal is inversely proportional to the number of influential authors. The light can be green, yellow, or red when the number of influencers posting on the same day is lower or equal to 10, 5, or 3, respectively.[14]

4.2 Analysis of abnormal returns

We finally check whether the algorithm based on social network analysis can adequately detect financially unstable days. In order to evaluate the accuracy of our algorithm, we develop an event study analysis following MacKinlay (1997). We measure whether abnormal returns occur for the stock discussed in the Reddit community right after the alert turns on.

The abnormal returns are constructed by defining an estimation window that goes from T_0 to T_1, and an event date τ. The event window goes from T_1 to $T_2 =$. $L_1 = T_1 - T_0$ and $L_2 = T_2 - T_1$ are respectively the length of the estimation and the event window.

[13] We do some robustness checks by changing the number of users with the highest in-degree centrality, and also modifying the parameters of the influencer set (window length and the number of critical users identified by the PageRank algorithm). The results are unaffected.

[14] These threshold numbers are not tuned in an in/out-of-sample analysis because of a limited number of time observations. Eventually, if the model is tuned might produce better performance in identifying future alerts. This analysis is beyond the scope of the paper and might be the topic for future research.

The abnormal returns are defined with the specification of the market model, to purge it by the market fluctuations:

$$AR_{\tau,s} = R_{\tau,s} - \hat{\alpha}_s - \hat{\beta}_s R_{\tau,m} \tag{4.8}$$

where $R_{\tau,s}$ is the return for security s at time τ, while $R_{\tau,m}$ is the market return. The parameters of the model are estimated via OLS over the estimation window $[T_0 : T_1]$. Under the null hypothesis, the abnormal returns, $AR_{\tau,s}$, are normally distributed with zero conditional mean and conditional variance $\sigma^2 (AR_{\tau,s})$ is set to be the variance of the OLS residuals $\sigma^2_{\varepsilon_t}$.

Under the null hypothesis, that event has no impact on the returns, the distribution of the abnormal returns in the event window is:

$$AR_{\tau,s} \sim N\left(0, \sigma^2 (AR_{\tau,s})\right) \tag{4.9}$$

The significance of the abnormal returns is tested via the non-parametric rank test by Corrado (1989). The abnormal returns are standardized as a ranked descending variable defined by:

$$K_{\tau,s} = \frac{\operatorname{rank}(AR_{\tau,s})}{1 + M_s} \tag{4.10}$$

where M_s are the number of observations in the sample for security s. The ranked variable is uniformly distributed in a [0,1] interval.
The variance computed across all the stock s observations is:

$$S^2_{\bar{K}} = \frac{1}{M_s} \sum_{t=0}^{T} (K_{s,t} - 0.5)^2 \tag{4.11}$$

where 0 and T stand for the first and last date of the sample. The significance is computed as a t-rank statistic for a standard uniform distribution with an expected value of 0.5:

$$t_{\text{rank}} = \left(\frac{K_{s,\tau} - 0.5}{S_K} \right) \tag{4.12}$$

These results can be used to make inferences on the absolute returns for every security in the event window.
We also test the impact of the detected events on the trading volume. Specifically, we

consider the abnormal trading volume computed as the volume traded on a given day divided by the average volume traded on the estimation window $[T_0 : T_1]$. For each day τ, the abnormal trading volume is defined as:

$$AVol_{\tau,s} = \frac{Vol_{\tau,s}}{\frac{1}{L_1}\sum_{t=T_0}^{T_1} Vol_{t,s}} \tag{4.13}$$

5 Results

This section consists of an empirical application of the methodology we design in the previous section. We present the days spotted by the alert tool and evaluate the associated abnormal returns with an event study analysis. For the latter analysis, we only consider the red-light alerts to leave out bias potentially generated during noisy days when many users publish on the platform.

5.1 Event detection

We run the alert system on six tickers: four meme-stocks (GME, AMC, TSLA, NOK) and two controls (AAPL and MSFT). The alert tool identifies 49 days where suspicious movements on the social network happen for GME, 10 for AMC, 12 for TSLA, and 4 for NOK, while only 2 and 1 for AAPL and MSFT, respectively. Our alert system can disentangle noise due to high-volume social activity and user coordination. For example, Reddit activity for AAPL is higher than both TSLA and NOK ones (Figures 1, 2), but the alert detects fewer warnings for AAPL due to the absence of coordination. The results, split by the severity of the alert, are shown in Table 1.

Tables 2 and 3 report all the dates and the users noticed by the alert system. The tool can trace the user acting like leaders for the movement. In Appendix B, we show a user network to highlight how agents interact on the social network.
In Figures 5, 6, 7, and 8 we report the daily time series of price and traded volume for the stocks under analysis, together with the days the alert system detects extraordinary activity and cooperation on the social network.

GME is by far the stock that attracted more attention and chatter on social media. Let us consider the short squeeze that happened at the end of January 2020 as

	Green	Yellow	Red	**Total**:
GME	20	3	26	49
AMC	5	2	3	10
TSLA	3	1	8	12
NOK	1	0	3	4
AAPL	2	0	0	2
MSFT	1	0	0	1

Table 1: Number of times the alert detects a potential influencer reporting the degree of intensity (green, yellow, red) of the alert.

the most striking episode of social network cooperation impacting the financial economy. We appreciate that our alert tool can detect abnormal and suspecting movements on the network many days before the social coordination affects the financial market. In addition, the tool can narrow down the few users that drive the movement, simplifying the eventual inspection by a surveillance authority.

Despite the lower media frenzy compared to GME, AMC, NOK and TSLA experience considerable attention on social media.

Unlike meme stocks, the financial performance of prices and volumes of non-meme stocks (AAPL and MSFT) is entirely unaffected by social media activity. Leading users ('fanatics') on social networks are aware that stocks with such high capitalization are challenging to implement pump-and-dump schemes or drive the price away from fundamental value. For this reason, the alert is rarely triggered and the identified alerting events are insignificant compared to those of meme stocks.

5.2 Analysis of the abnormal returns and volumes

The submissions by the influencers during the days of unusual activity, detected by the two stages of the alert system, are considered events potentially triggering a response in the financial market. We analyze abnormal returns constructed with a market model to evaluate whether the submission from the influencers triggers a reaction in the financial markets. For each detected event, i.e., when the alert system turns on ($\tau = 0$), we compute the abnormal return on the estimation window [-41:-11], of length $L_1 = 30$

trading days and we consider as event window the period [-10:10], of length $L_2 = 21$ trading days. The abnormal return is defined with a market model, specified in (4.8), with the CRSP index as a proxy for the market returns. We impose a minimum of 10 days between two events to avoid contagion on the event window and consider the events independent. If an event happens on a non-trading day, we consider the next trading day as the event date.

Under these conditions, we end up with 27 events for GME, 3 events for AMC, 7 events for TSLA, and 3 for NOK over the sample period.

Considering that Reddit users are mostly inexperienced retail investors, we assume they primarily trade with a long position on the stock. Indeed, their primary concern is to move the price up by massively buying the stock. For this reason, we report and evaluate the significance of the event that generates the highest positive abnormal returns in the ten days after. Results are reported only for meme stocks[15].

[15]Notice that we do not report results for non-meme stock because the alerts detected are very few, see Table 1.

Day (τ)	Abnormal returns	Abnormal volumes
-10	0.034	0.587*
-9	0.001	-0.964
-8	-0.1241	-0.750
-7	-0.072	0.434*
-6	-0.004	-0.621
-5	0.007	0.052
-4	0.019	1.031*
-3	0.073*	0.416
-2	0.194*	-0.210
-1	0.036	0.087
0	0.001	-0.604
+1	**0.339****	-0.381
+2	-0.0002	**2.130****
+3	-0.207	0.380
+4	0.153*	0.684*
+5	-0.147	-0.309
+6	-0.077	0.088
+7	0.011	0.042
+8	0.068*	-0.849
+9	0.069*	-0.258
+10	-0.039	-0.208

Table 4: Average abnormal returns and volumes for a [-10:+10] days event window around the event day. ***, **, * represents significance at 1%, 5% and 10%. The highest value is in bold.

Table 4 reports the average abnormal returns and volumes at different horizons, with respective significance levels. Significant positive abnormal returns are detected around the alert day, with higher abnormal returns the day after ($h = +1$). In Figure 9, we provide the graphical representation of the average abnormal returns/volumes and cumulated average abnormal returns/volumes for the meme stocks (GME, AMC, TSLA, NOK). The average abnormal return presents higher volatility after the event, with a significant positive impact the day after. A similar pattern is followed by abnormal volumes.

In Figure 10 we provide the percentage of positive abnormal returns at different horizons for the meme-stocks. The plot could be described as follows: for each ticker, we compute the rate of positive abnormal returns four, three, two, and one day before the event: $h = -4, -3, -2, -1$; the percentage of positive abnormal returns during the event day $h = 0$; and the percentage of positive abnormal returns one and two days after the event: $h = +1, +2$. If the histogram for a stock at $h = 0$ is 0.5 means that during all the alert days, there are half-positive and half-negative abnormal returns for the stock in consideration. Therefore, the plot shows how many positive abnormal returns are generated around the alert days. Instead of focusing on the overall volatility of stock around the event dates, with this plot we can evaluate the portion of *positive* abnormal returns for each stock. By looking at the figure, we generally notice an upward trend for the rate of positive abnormal returns from days before the event to days after the event.

Overall, the event analysis suggests significant abnormal returns following the alert dates for the meme stock. The alert for non-meme stocks turns on rarely and without generating abnormal returns.

6 Conclusions

Social media are a powerful and impacting tool to disseminate information and stir a vast mass of people. This paper provides empirical evidence of the echo chamber effect on the financial market from the social network, a form of market manipulation (albeit indirect). Reddit and, in general, social media are great places to share advice and manipulate poorly-informed, unsophisticated, and prone to be convinced investors.

We provide evidence that influential social media users, or to use a word a là Pedersen, the fanatics can coordinate many naive investors to provoke the desired stock price movement. The fanatics can effectively undermine the financial market stability by persuading inexperienced and easily reachable people.

We design an alert system to detect abnormal movement related to a specific stock on social media based on extraordinary activity in terms of volume and the detection of a potential manipulator that coordinates the mass movement.

While it is far from our duty to evaluate whether the promotion and persuasion practice falls within the boundaries of the law, our consideration concerns the market microstructure models. In front of these episodes, the retail investors can no longer be relegated to a residual category of 'noise traders', but models should consider that many small and apparently harmless investors if aggregated and coordinated, can generate a disruptive effect on the financial market.

In the end, the entire analysis spots significant differences between the meme and non-meme stocks. The detection system finds alerts for both categories. Still, they never turn into abnormal returns for the non-meme stocks, suggesting fewer chances of suspicious trading activity or market manipulation in those cases. Moreover, we find evidence of positive significant average abnormal returns around the days of the alert and generally high and persistent volatility, especially the days. Interestingly, social media are a continuously evolving phenomenon, and already from the descriptive statistics, we notice differences in the behavior of Reddit users when the sample period is extended. Direct participation in the stock discussion for a particular stock decreased after the summer of 2021. The behavioral changes of users might compromise the effectiveness of our tool. For this reason, it is crucial to constantly update the monitoring techniques evaluating potential breaks in the behavior of social network agents.

All in all, future works should focus on two main aspects. Firstly, our tool should be backtested by using more data in terms of a larger cross-section of stocks and more time observations. The backtest should also consist in tuning parameters of the algorithm (for example by finding the optimal thresholds for the alert to turn on). Secondly, an additional layer of analysis should focus on applying NLP techniques to the text of Reddit users and to the analysis of data from other social networks (for example Twitter).

Appendices

A Data download

For each tree, we have as many rows in the data frame as the number of comments, and each row contains the following information:

- `title`: the textual content of the initial submission;
- `body`: the textual content of the comment;
- `name`: the id of the author of the comment (each id is prefixed by `'t1_'` to specify the author made a comment activity);
- `parent_id`: the author of the parent comment to which the comment in question refers to (the `parent_id` can be prefixed by `'t1_'` if the author of the comment replies to an other comment or it can be prefixed by `'t3_'` if the author of the comment replies to the top-level post, i.e. the submission);
- `author_name`: the name of the author who post the initial submission;
- `depth`: the level of the comment tree at which the comment in question is located (if a tree is composed by the initial submission only, the depth is 0; if the comment refers to the initial submission the depth is 1; if the comment refers to a comment in the first level, the depth is 2; and so on);
- `score`: the number of up-votes minus the number of down-votes obtained by the comment;
- `score_submission`: the number of up-votes minus the number of down-votes obtained by the initial submission;
- `upvote_ratio`: the percentage of upvotes on the total votes received by a submission;
- `time_submission`: date and time at which the initial submission is published;
- `time_comment`: date and time at which the comment is published;

- `num_comment`: number of comments below the initial submission that compose the tree;
- `flair`: a tag used to categorize the post according to the topic it deals with; they are subreddit specific and in the case of the subreddit *r/WallStreetBets* the users can select among the following ones:
  - YOLO, the acronym for 'You Only Live Once, it can be used for posts presenting extremely aggressive investment strategies with a consistent value at risk;
  - DD, the acronym for Due Diligence, must be applied to post presenting research on a specific company/sector/trade. It should include sources and citations;
  - Discussion, an idea or article that you would like to talk about;
  - Gain, to show off a solid winning trade;
  - Loss, to show off a brutal, crushing loss;
  - Earnings Thread, weekly earnings discussion thread or a specific earnings event;
  - Daily Discussion, daily catch-all thread for discussions;
  - Mods, only for official business.
- `distinguished`: if a bot automatically performs the commenting activity, the variable reports the wording 'Moderator', none otherwise, when a non-automatic user adds the comment.

Note that in the case of submission without comments below, the data frame has a single row with empty values for the comments-related variables.

B The Social Network of Reddit users

In Figure 11, we present an example of network user graph on January 31st, 2021, where the main submissions contain the ticker AMC. There are 15.534 users (represented by the nodes) interacting among them on the platform throughout commenting activity (the 21.032 edges connecting them).

The directed edges point from the comment's author to the author of the main submission. The peripheral nodes in the graph are the less connected users; in the central part of the network, the most connected and central users: the colored ones are the users with the highest in-degree centrality.

Figure 12 presents the network graph for AAPL a and MSFT on two alert dates, respectively June 22nd, 2021 for AAPL and on May 20th, 2021 for MSFT. Compared to the meme-stock case, the non-meme-stocks present a feeble activity on the social network even in extraordinary occasions that determine the alert activation.

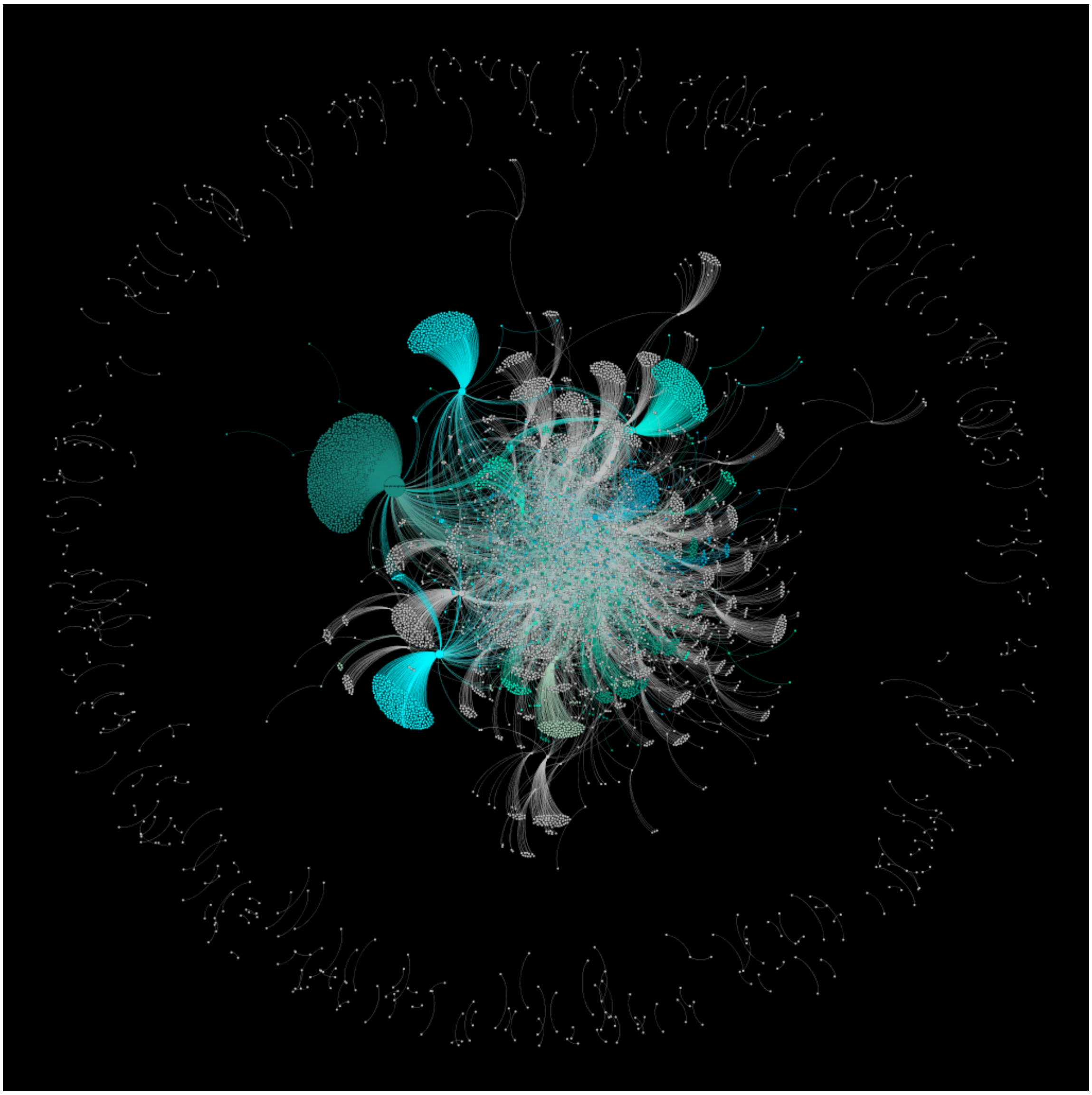

Figure 4: The network of users interacting on Reddit on January 14st, 2021. The network reports the interactions of users posting a submission containing the wording 'GME'. The network has 6 465 nodes and 8 741 edges. The top 10 nodes with the highest in-degree centrality are colored in blue.

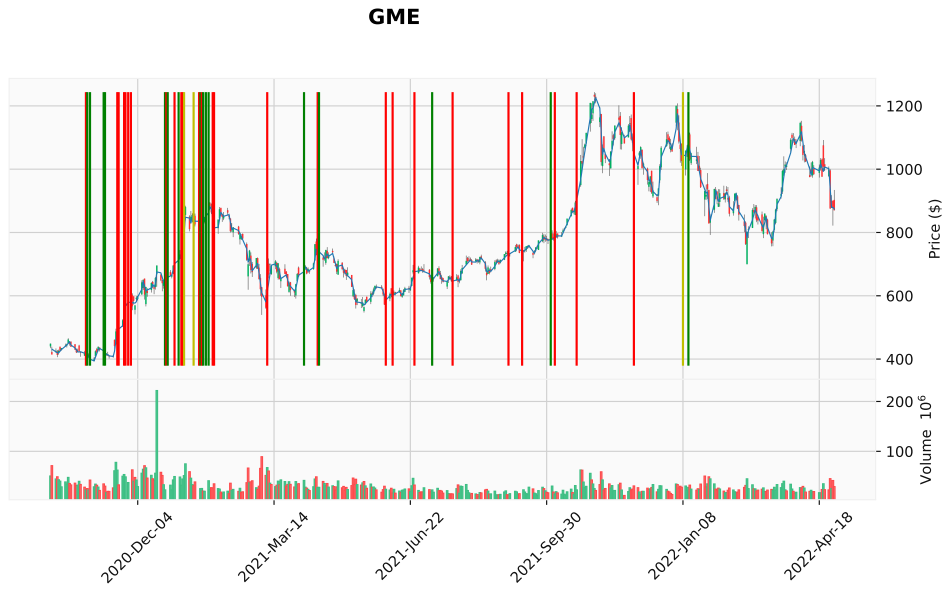

Figure 5: The figure shows financial data together with the alert days for the stock GME over the period October 2020 - June 2021. The above panel of the figure presents the time series of the daily price with a candlestick chart and the vertical blue lines are the days when the alert system turns on; in the bottom panel, the daily traded volume over the corresponding period.

Figure 6: The figure shows financial data together with the alert days for the stock AMC over the period October 2020 - June 2021. The above panel of the figure presents the time series of the daily price with a candlestick chart and the vertical blue lines are the days when the alert system turns on; in the bottom panel, the daily traded volume over the corresponding period.

Figure 7: The figure shows financial data together with the alert days for the stock TSLA over the period October 2020 - June 2021. The above panel of the figure presents the time series of the daily price with a candlestick chart and the vertical blue lines are the days when the alert system turns on; in the bottom panel, the daily traded volume over the corresponding period.

Figure 8: The figure shows financial data together with the alert days for the stock NOK over the period October 2020 - June 2021. The above panel of the figure presents the time series of the daily price with a candlestick chart and the vertical blue lines are the days when the alert system turns on; in the bottom panel, the daily traded volume over the corresponding period.

GME

Alert date (τ)	Influencer(s)
2020-10-27	SomebodyCallJaRule (**red**)
2020-10-28	neothedreamer (green)
2020-10-30	neothedreamer (green)
2020-11-09	DeNovaCain (green)
2020-11-10	Analoghogdog (green)
2020-11-19	Ackilles (green), monaliza24, imboredsoyh (red)
2020-11-20	imboredsoyh (green), neothedreamer (red)
2020-11-24	Ackilles (red)
2020-11-25	anon10384 (red)
2020-11-27	SIR_JACK_A_LOT (red)
2020-11-29	MadeTheAccountForWSB (green)
2020-12-24	robbinhood69 (red)
2020-12-25	danielbauwens (green)
2020-12-26	Uberkikz11 (red)
2020-12-31	maxfort86 (green)
2021-01-03	ShortTheNasdaq, Uberkikz11 (red)
2021-01-05	jbro12345 (red)
2021-01-06	SHoo98 (yellow)
2021-01-07	SHoo98 (yellow)
2021-01-14	DeepFuckingValue (red)
2021-01-18	its-Loki (green)
2021-01-19	gardeeon (green), DeepFuckingValue (red)
2021-01-20	Unlucky-Prize (green)
2021-01-21	Unlucky-Prize (green)
2021-01-23	Unlucky-Prize (green)
2021-01-25	DeepFuckingValue (red)
2021-01-28	DeepFuckingValue (red)
2021-01-29	DeepFuckingValue OPINION_IS_UNPOPULAR (red)
2021-03-09	dumbledoreRothIRA (green)
2021-04-05	OPINION_IS_UNPOPULAR (red)
2021-04-15	FatAspirations (green) OPINION_IS_UNPOPULAR (red)
2021-04-16	DeepFuckingValue (red)
2021-06-04	irishdud1 (red)
2021-06-09	alphaomega_2021 (green)
2021-06-25	Chillznday (red)
2021-07-08	karasuuchiha (red)
2021-07-23	nobjos (red)
2021-09-02	chayse1984 (green)
2021-09-12	cbass37, Agreeable_Ingenuity8 (red)
2021-10-03	Environmental_Ad222 (red)
2021-10-06	karasuuchiha (red)
2021-10-22	luckystrike921 (yellow)
2021-12-03	karasuuchiha (green)
2022-01-08	HollaAtYaYOLO (green)
2022-01-12	LavenderAutist (red)
Alerts: # green = 20, # yellow = 3, # red = 26	

AMC

Alert date (τ)	Influencer(s)
2020-10-18	jimbofbx (green)
2020-11-09	Killtrend (green)
2021-01-31	dhiral1994 (yellow) BrandinoGames (green)
2021-02-25	Stockgandu (green)
2021-04-16	TheNovaeterrae (**red**)
2021-05-30	nobjos (**red**)
2021-07-07	Chillznday (**red**)
2022-04-11	'Head-Instruction-195 (**green**) DreamimgBig (yellow)
Alerts: # green = 5, # yellow = 2, # red = 3	

Table 2: The Table reports for GME and AMC stock the events spotted by the alert system. For each detected event, it indicates the date, the user(s) under investigation, and the light of the alerts.

TSLA

Alert date (τ)	Influencer(s)
2020-11-09	xxx69harambe69xxx (**green**)
2020-11-14	hallalex831 (green)
2020-11-17	ChefBaconz (red)
2020-12-24	ninkorn (red)
2021-04-01	TU_NYCE (yellow)
2021-04-06	TU_NYCE (red)
2021-05-07	NrdRage (red), neothedreamer (red)
2021-06-03	ElkFalse6637 (red)
2021-06-24	Chillznday (red)
2021-12-02	pgrdub (green)
2021-12-02	jjd1226 (red)
Alerts: # green = 3, # yellow = 1, # red = 8	

NOK

Alert date (τ)	Influencer(s)
2020-10-27	sykotikqp (red)
2021-01-21	No_Hat8155, Character_Repair661 (red)
2021-03-15	wallstreetbagel (**red**)
2021-06-04	Hot-Room-240 (green)
Alerts: # green = 1, # yellow = 0, # red = 3	

Table 3: The Table reports for TSLA and NOK stock the events spotted by the alert system. For each detected event, it indicates the date, the user(s) under investigation, and the light of the alerts.

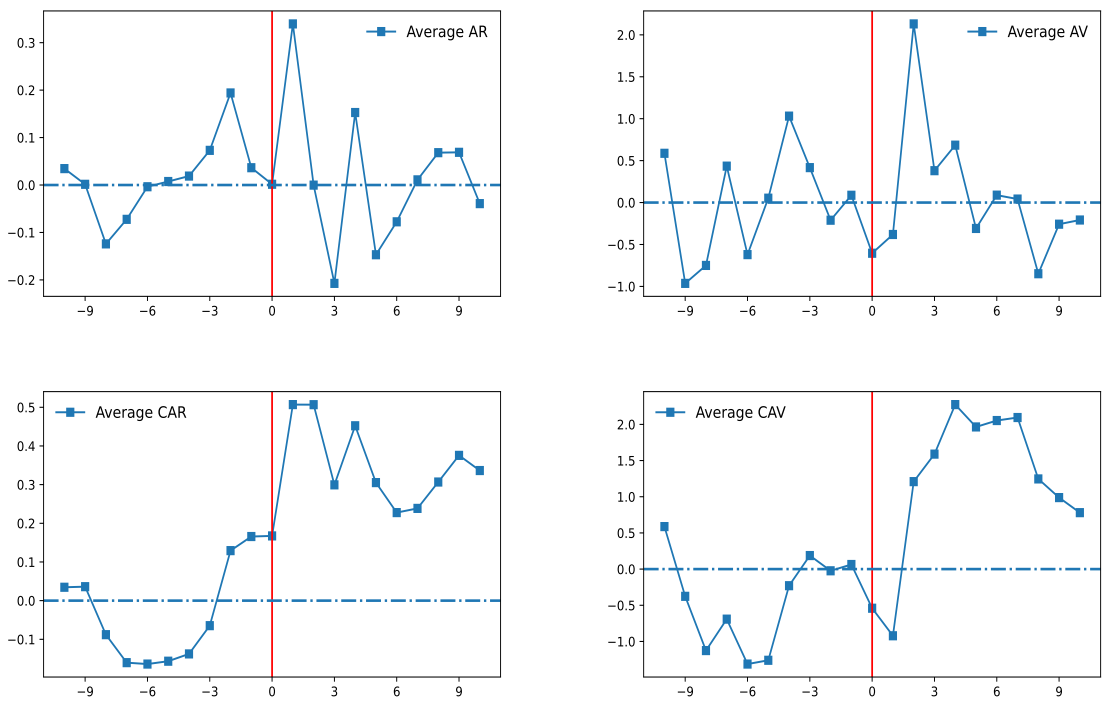

Figure 9: Average abnormal returns/volumes (top panel) and cumulated average abnormal returns/volumes (bottom panel) over the whole period [-10,+10].

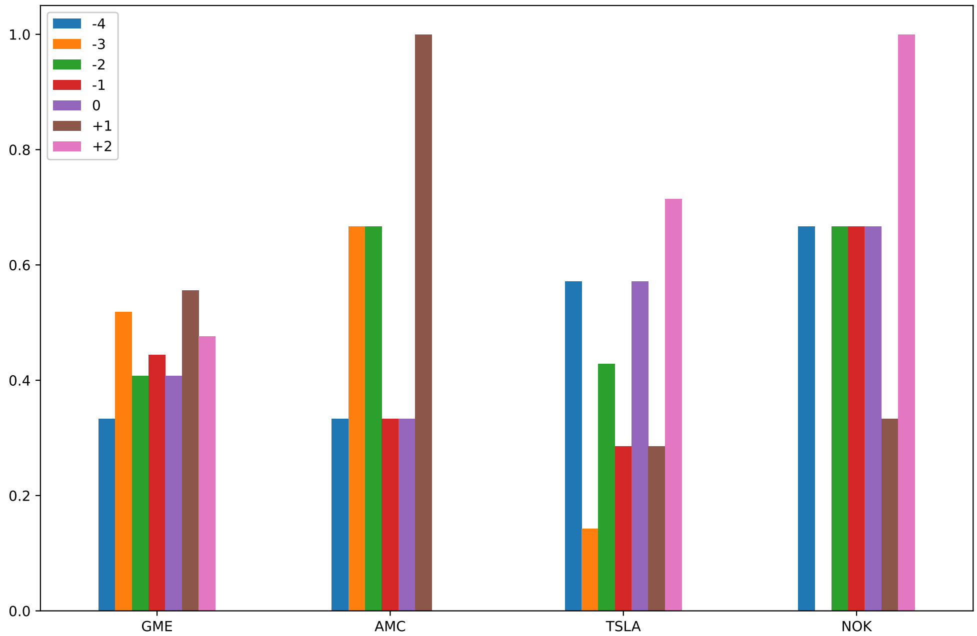

Figure 10: Percentage of positive abnormal returns at $h = \{-4, -3, -2, -1, 0, +1, +2\}$ for the meme-stocks.

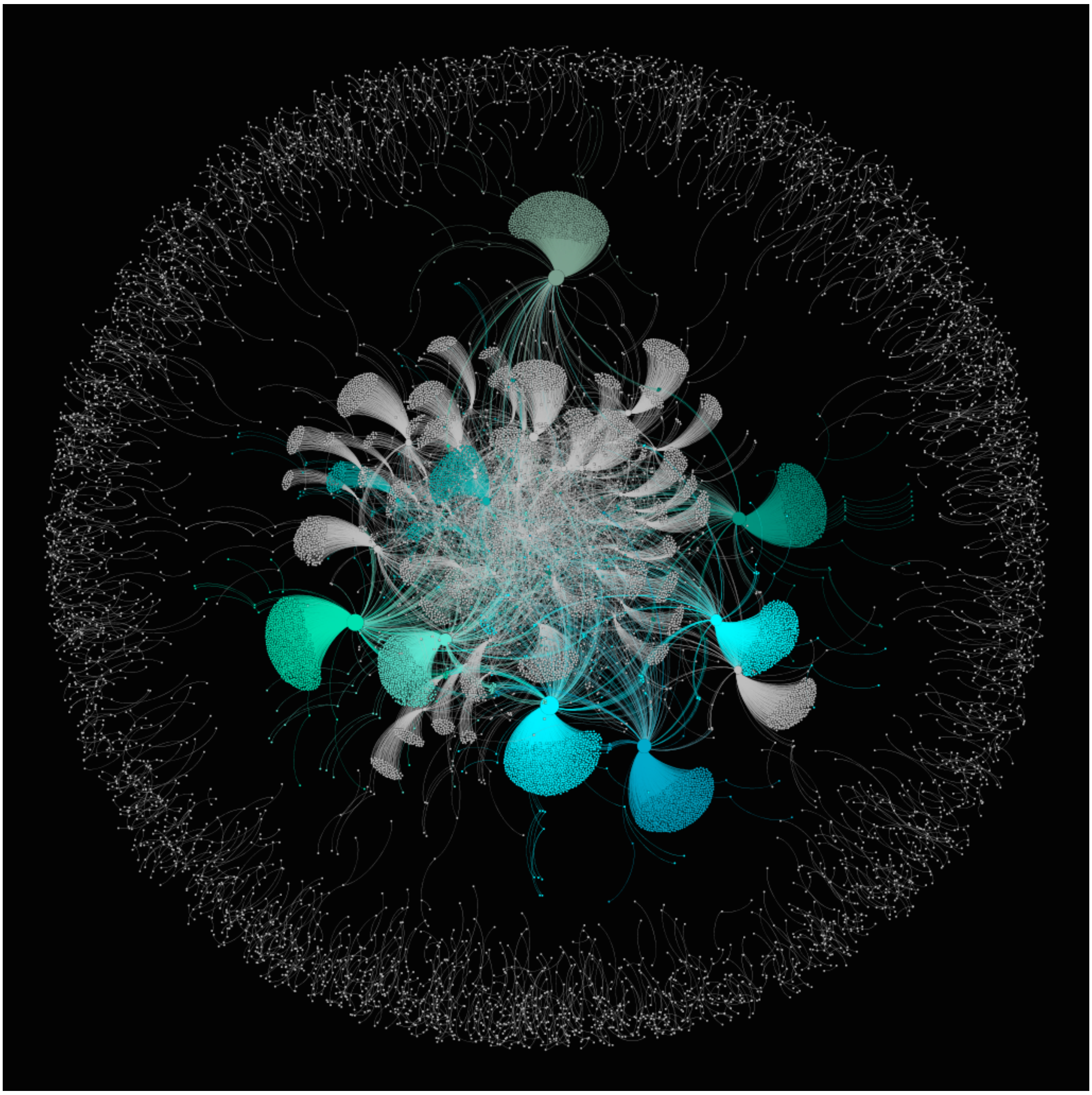

Figure 11: The Figure shows the network of users interacting on Reddit on January 31st, 2021. The network reports the interactions of users posting a submission containing the wording 'AMC'. The graph contains 15.534 nodes and 21.032 edges. The colored part of the network is the nodes involved in the net of the 10 users with the highest in-degree centrality.

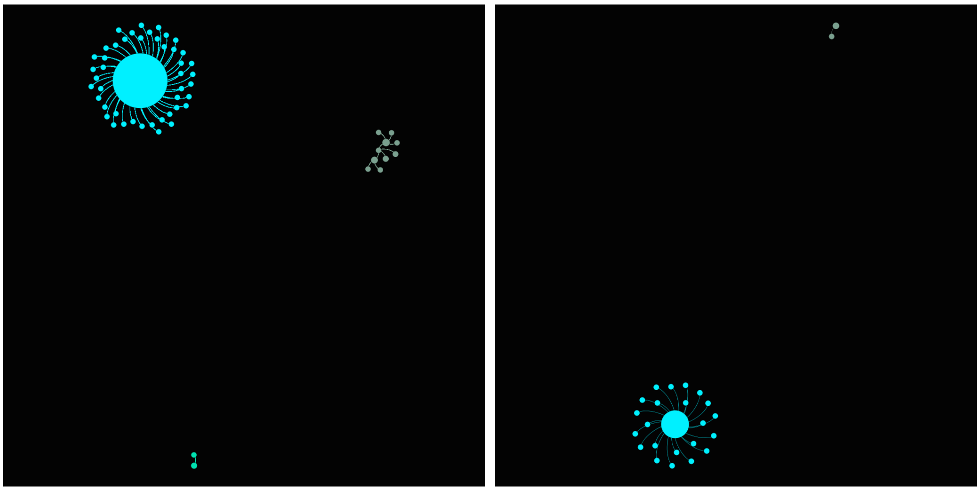

Figure 12: The Figure shows the network of users interacting on Reddit on June 22nd, 2021 for AAPL and on May 20th, 2021 for MSFT. The network shows the interactions of users posting a submission containing the wording 'AAPL' and 'MSFT' respectively.

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